Dhruv Ramani

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Education

Newton School of Technology (Rishihood University) India
B.Tech in Computer Science and Artificial Intelligence (3rd Year) *Aug 2024 – 2028*

- **Relevant Coursework:** Neural Networks and Deep Learning, Generative AI, Linear Algebra, Statistics, Data Analytics and Visualization

Projects

DriftLens – Reward-Hacking Monitoring for RL *(ongoing)* | *PyTorch, SB3, Gymnasium* Sep 2026 – Present

- Instrumented PPO to log per-update telemetry (anchor-gradient alignment, effective rank and subspace angles of parameter updates, behavioral entropy) and built two Gymnasium benchmarks, `ShortcutGrid` and `TippingGrid`, with deliberately exploitable proxy rewards and a separate true-task oracle kept out of the online score.
- Ran multi-seed studies with predeclared gates, frozen thresholds and untouched test layouts, and reported the negative results: the composite score false-alarmed on **3/3 normal controls**, and a proposed benchmark condition failed its AUC ≤ 0.60 gate at **0.731**.
- Built a semantic-twin control: **5/5 paired policies** matched on all 392 learner-visible updates while differing on the hidden objective, showing that proxy-only signals cannot identify reward hacking without trusted outcome audits.

Rural Health Triage – OpenEnv RL Environment | *FastAPI, Pydantic, Docker* Apr 2026

- Built an OpenEnv-compliant environment in which an agent triages synthetic patients across **9 WHO-guideline disease presentations**: asking symptoms, ordering tests under a budget, then diagnosing, treating, or referring.
- Designed **3 graded tasks** (budget \$20 → \$8) with deterministic graders (ground truth fixed at case generation, no LLM judge) and per-step shaped rewards that penalize dangerous treatment and discharging critical patients.
- Benchmarked a Llama-3.1-8B agent at **0.534** mean score: it identified diagnoses but scored zero on treatment selection, a capability gap the task-level reward breakdown isolates.

Mavis – Real-Time CV Fitness Coach | *MediaPipe, OpenCV, React, TypeScript* Nov 2025 – Present

- Engineered an on-device pose pipeline (33 MediaPipe landmarks → EMA smoothing → stillness-gated calibration → per-exercise state machine) that counts reps and scores each out of 100, with body-relative fault detectors so a stance shift is not misread as a form fault. No video leaves the device.
- Cut the false-fault rate from **61%→5%** (shoulder press) and **46%→10%** (bicep curl) by setting every tolerance from measured distributions over 128 recorded reps; a pytest suite replays landmark fixtures to assert clean reps pass and poor form is still caught.

Exoplanet Detection Pipeline | *Scikit-learn, XGBoost, FastAPI, Next.js* Feb 2025 – Present

- Built an end-to-end transit-detection pipeline over live NASA TESS data (MAST lightcurves, `wotan` detrending, Transit Least Squares) with five physics vetting checks and a plain-English reasoning trail for every verdict.
- Trained a Platt-calibrated RandomForest + XGBoost ensemble on 7,325 Kepler objects: **0.957 ROC-AUC, 0.917 PR-AUC** on 1,465 unseen targets.
- Authored a preprint quantifying **Kepler→TESS domain shift**: ROC-AUC falls 0.96 → 0.72 and recall recovers 0.28 → 0.72 via a cross-validated threshold, validated with injection–recovery maps and a label-permutation null test.

Technical Skills

Languages: Python, JavaScript/TypeScript, Dart, SQL
RL / ML: PyTorch, Gymnasium, Scikit-learn, XGBoost, model calibration
Web & Mobile: React.js, FastAPI, Node.js, Tailwind CSS
Libraries & Tools: NumPy, Pandas, OpenCV, MediaPipe,, Docker, pytest